

Boundary Work in Knowledge Teams

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The purpose of this article is to promote an open systems perspective on team research. The authors develop a model of team boundary activities: boundary spanning, buffering, and reinforcement. The model examines the relationship between these boundary activities and team performance, the moderating effects of organizational contextual factors, and the mediating effect of team psychological safety on the boundary work–performance relationship. These relationships were empirically tested with data collected from 64 software development teams. Boundary spanning, buffering, and boundary reinforcement were found to relate to team performance and psychological safety. Both relationships are moderated by the team's task uncertainty and resource scarcity. The implications of the findings are offered for future research and practice.

Keywords: boundary work, boundary activities, team effectiveness, software teams

Today, work teams must increasingly operate in organizations that are becoming debureaucratized (Denison, Hart, & Kahn, 1996), boundaryless (Kerr & Ulrich, 1995), network based (Majchrzak, Jarvenpaa, & Hollingshead, 2007), temporarily structured (Bechky, 2006), geographically dispersed, and electronically mediated (Gibson & Gibbs, 2006; Kirkman & Mathieu, 2005). This profound change away from formal structures and bureaucratic organizing principles raises important questions as to how teams manage boundaries in an organizational environment wherein cross-functional teams are now prevalent (Cappelli, 1999) and members routinely participate in multiple teams (Marks, DeChurch, Mathieu, Panzer, & Alonso, 2005). Moreover, work teams increasingly bring together members with differentiated specialization to work on complex problems, with the expectation that there will be extensive knowledge creation and transfer across team, organizational, and even interorganizational boundaries (Argote, McEvily, & Reagans, 2003; Majchrzak et al., 2007; Reagans & Zuckerman, 2001).

Team boundary work, defined as the activities that a team engages in to establish and maintain boundaries and manage interactions across those boundaries, has increasingly devolved from the management hierarchy to the team itself. The essential problem for a team is to create boundaries that are porous enough to allow resources and information in but resistant enough to avoid uncertainty about who is on the team and whether these members

are accountable for its collective outcomes (Hackman, 2002). Therefore, team boundary work includes acquiring information and resources and managing relationships with external stakeholders, as well as protecting team resources (including members' time and energy) from competing external demands (Ancona & Caldwell, 1988; Hansen, 1999; Reagans & Zuckerman, 2001). In addition, teams must maintain internal boundaries to engender members' sustained commitment to the task at hand (Druskat & Wheeler, 2003; Hirschhorn & Gilmore, 1992; Hirst & Mann, 2004; Victor & Stephen, 1994).

Researchers working in recent years have generally agreed on the informational and resource benefits that crossing boundaries brings teams (e.g., Ancona & Caldwell, 1992a; Argote et al., 2003; Cross & Cummings, 2004; Hansen, 1999). But there is little agreement as to the dimensions of boundary work. Furthermore, the impact of team-level boundary work is not fully understood in terms of either antecedents or consequences (Golden & Veiga, 2005). And there are drawbacks: Teams that engage in excess external orientation may face role overload and reduced team viability (Marrone, Tesluk, & Carson, 2007). Thus, there is a need to elaborate on the nature of the generally observed association between team boundary work and performance and to establish if this association holds for different tasks and organizational environments.

In the present study, we investigate these important questions. First, we advance a model that conceptualizes several different types of boundary work—boundary spanning, buffering, and reinforcement—and tease out how these boundary activities collectively and separately contribute to team performance and psychological safety. Second, we investigate the role of contextual factors—task uncertainty and resource scarcity—in moderating the relationship between boundary work and team performance. By bringing in boundary work and environmental contingencies, we intend to further an open systems approach to team studies, which have been dominated by closed systems approaches (Ancona, 1990; Choi, 2002). The theoretical model was tested with data collected from 64 software development teams in a leading information tech-

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nology firm that specializes in developing business software for both commercial and internal clients.

Theoretical Framework

Research on team boundaries falls into two major traditions. One group of researchers takes an external perspective, viewing teams through an open systems lens and focusing on issues of demarcation with environment and system frontiers (e.g., Scott, 1998). A second group of researchers adopts a teamwork perspective based on small group research, emphasizing the importance of processes for team effectiveness (e.g., Marks, Mathieu, & Zaccaro, 2001). Both perspectives provide unique insights into the issue of team boundaries and are explored in greater detail below.

For almost a quarter century, the external perspective on work teams has emphasized the importance of boundaries and the necessity of managing them. This approach has extended the internal focus of traditional team research (Ancona & Caldwell, 1988; Choi, 2002; Gladstein, 1984; Scott, 1998). The impetus for an external perspective on teams came from studies of research and development (R&D) and the diffusion of technological innovations. These studies find that team boundary spanning activities play a key role in gaining the team access to needed information (Allen, 1984; Tushman, 1977; Zmud, 1983). Recognizing the open system nature of organizational teams, Ancona and Caldwell conducted a series of studies of the external activities of teams, using interview and questionnaire methods (Ancona, 1990; Ancona & Caldwell, 1988, 1990, 1992a). In these studies, they identify the external roles adopted by members of new product development teams and classify them into team boundary strategies.

Ancona and Caldwell's pioneering work was followed by attempts to reexamine or extend the external activities model. Field studies of software development teams (Guinan, Coopridge, & Faraj, 1998) and new product development teams (Keller, 2001) have found a positive link between the degree of a team's boundary spanning and the stakeholders' rating of the team's performance. Similarly, Druskat and Wheeler (2003) conducted an in-depth investigation of self-managed teams at a manufacturing plant, finding that undertaking boundary spanning activities strongly affects team performance. These activities include building relationships, scouting for information, and persuading others to support the team's work. In an in-depth case study, Cross, Yan, and Louis (2000) traced changes in patterns of boundary activity associated with restructuring from a functional to a team-based organization.

The authors of these studies clearly recognized the importance of team boundary work and the external roles played by team members. Thus, this external view shows that the effectiveness of a team is a function of both its internal characteristics and processes and contextual factors determined by the larger organization and environment within which the team operates (Hyatt & Ruddy, 1997). However, work on team boundaries has remained sporadic and unsystematic. Rigorous theoretical and empirical work is needed in several salient areas. First, although boundary activities have been examined at the team level, these studies have not gone beyond the traditional boundary functions observed at the organizational level (i.e., boundary spanning and buffering; Scott, 1998). The literature is silent on how teams maintain and reinforce their boundaries over time. Second, the observation that some external

activities are beneficial whereas others hinder performance (Ancona & Caldwell, 1992a; Hirst & Mann, 2004; Keller, 2001) implies the need for a refined effort to delineate the effect of each particular type of boundary work on team effectiveness. Third, because teams face different tasks and operate in dissimilar environments, there is a growing need to theorize boundary work models that incorporate the relevant organizational and environmental contexts. In this article, we propose such a model of team boundary work.

In recent years, the study of work teams has undergone a marked shift away from traditional input-process-output models, which emphasize what makes certain teams more effective than others, to a focus on "mediating processes that explain why certain inputs affect team effectiveness and viability" (Ilgen, Hollenbeck, Johnson, & Jundt, 2005, p. 519). This has led to an increased focus on mediating factors and an important conceptual clarification of what was previously used as a catch-all process label. Marks et al. (2001) argued that processes, which they defined as interdependent team activities that orchestrate task work, should be distinguished from emergent states—namely, the evolving cognitive, emotional, and affective qualities of teams. This conceptual clarification provides an important distinction between process activities, such as boundary work, and emergent state constructs, such as psychological safety or team empowerment (Mathieu, Gilson, & Ruddy, 2006).

Indeed, current research is highlighting the benefits of specific cross-boundary interactions. In an experimental study of 46 student teams in which cross-team interdependence was manipulated, teams that engaged in quality cross-team interaction were found to perform better under conditions of increased interdependence than teams in a low interdependence situation (Marks et al., 2005). In a similar vein, field studies of consulting and R&D teams have indicated that specific cross-boundary interactions provide an effective means of acquiring information necessary for team performance (Cross & Cummings, 2004; Hansen, 1999; Marrone et al., 2007; Reagans & Zuckerman, 2001). In all of these studies, the cross-boundary interactions and team processes, not just the internal team processes, are found to impact team performance, which may confirm Choi's (2002) suggestion that internal and external team processes are not necessarily supplementary and may be synergistic.

Overall, the scholarly literature on boundary work has retained its focus on identifying specific dimensions of boundary work that may be linked to team performance (e.g., Ancona & Caldwell, 1992a). We suggest that such an approach, although appropriate for teams engaged in complex R&D tasks that necessarily require a continued flow of resources and feedback to important stakeholders, can be complemented by a more holistic view of boundary work based on a recognition of the multiple roles played by the boundary itself. We build on Yan and Louis's (1999) conceptualization of boundary work in organizations. Taking an open systems perspective, these authors actively recognized how the changes in organizational structure away from functional chimneys and the organizing principles of bureaucracy have led to increased demands on organizational members and teams. As boundaries become more permeable, teams face increased demands on members, including burdens imposed by other teams that the members serve on. Teams must span their own frontiers to access critical resources and dispose of their output and must buffer themselves

from the disruptive demands imposed by the external environment. In addition, teams must continually reinforce their boundaries to demarcate their space and establish their identity and distinct mission.

For the purposes of this article, we define team performance narrowly, as the stakeholder rating of team output. But because we recognize that performance may include outcomes that sustain team members' ability to work with each other (e.g., Hackman, 2002), we also focus on psychological safety as an important indicator of the quality of the intrateam environment. Psychological safety, defined as a belief that others will respond positively to questions, mistakes, or feedback seeking, provides an indication that members are engaged in behavior supportive of innovation and learning (Edmondson, 1999). In the balance of this section, we identify three distinct types of boundary work: boundary spanning, boundary buffering, and boundary reinforcement. We then articulate how each of these categories affects team performance.

Boundary Spanning

Organizational teams seldom work free from external scrutiny and seldom have all of the resources they need within their boundaries. Boundary spanning refers to actions a focal team undertakes to reach out into its environment to obtain important resources and support. Teams and organizational units are often involved in activities such as bargaining and negotiation, contracting and cooptation, and alliance and coalition building. To do this, they engage in various boundary spanning strategies (Scott, 1998). Perspectives on boundaries as frontiers for transactions among systems (Katz & Allen, 1985) and as interfaces between systems and their environment (Friedlander, 1987; Thompson, 1967) represent foundations for the concept of spanning as a key type of boundary work.

Previous studies of boundary spanning provide insights relevant to team performance. Ancona and Caldwell (1988, 1992a) found that the external activities teams engage in are associated with team outcomes. Effective product development teams have "ambassadors," "task coordinators," and "scouts," all of whom work with contacts in the larger organization within which they operate to acquire the information, resources, and support they need to carry out the task at hand. Moreover, Ancona and Caldwell observed that such groups reach beyond the organization to map trends such as technical developments relevant to the group's activities and influence critical external players. Their empirical evidence suggests that ambassadorial and task coordinator activities are strongly associated with team performance, whereas scouting activities are negatively associated with performance. In a study of 66 software development teams, Guinan, Coopride, and Faraj (1998) tested a model of team performance that includes both internal and external team processes and found that championing activities is positively related to stakeholder-rated performance. Similarly, Keller (2001) found that the amount of external task-related communications in research and new product development teams is positively related to stakeholder ratings of team performance. More recently, in a study of self-managed teams at a manufacturing plant, Druskat and Wheeler (2003) reported that undertaking boundary spanning activities strongly affects team performance. These activities include building relationships,

scouting for information, and persuading others to support the team's work, according to these researchers.

To summarize, through boundary spanning, teams reach out to detect internal and external demands, secure needed resources and support to respond to these demands, promote the team's work, and build goodwill among stakeholders. These activities help the team accomplish its objectives, thereby contributing to team performance.

Hypothesis 1: Boundary spanning is positively associated with team performance and psychological safety.

Boundary Buffering

In contrast to spanning, which is a strategy of engagement, buffering is a strategy of disengagement. In buffering, an entity closes itself off from exposure to the environment. An entity buffers to protect itself against uncertainties and disturbances from without, thereby enhancing the possibility of rational action within (Scott, 1998). Perspectives that view organizational boundaries as demarcations or perimeters emphasize boundary-buffering functions (Miller, 1967; Thompson, 1967). Researchers adopting this approach argue that a system must build and maintain well-defined boundaries that protect the system from interference as a condition of its existence. Buffering, as an action that protects core processes, may be undertaken either in response to or in anticipation of disruptive forces within the environment.

In debureaucratized organizations, members are increasingly the target of demands external to the team, many of which are mutually exclusive and thus in competition for members' time, attention, energy, and commitment. On the basis of a survey of *Fortune* 1000 corporations, Mohrman, Cohen, and Mohrman (1995) reported that knowledge workers are simultaneously involved in multiple teams, "a stretch that often caused conflicting priorities and worked against the effectiveness of the teams" (p. 256). Under such conditions, a team has to engage in substantial boundary buffering. Evidence of buffering includes the use of both formal strategies and procedures and informal codes and norms for deflecting and managing external demands on team members, along with other types of outside pressure or interference. Ancona and Caldwell (1988) found that, empirically speaking, the functioning of "sentries and guards" epitomizes buffering tactics of information and resource control, which are a means of defense against external agents penetrating the group. Although guard activities in the form of limiting interaction and avoiding releasing information were perceived negatively by external stakeholders, these activities were positively received by team members. In summary, buffering strengthens the team's boundaries against external interference and creates a protected internal atmosphere for the members, thereby contributing to team performance.

Hypothesis 2: Boundary buffering is positively associated with team performance and psychological safety.

Boundary Reinforcement

Boundary spanning and boundary buffering are relatively well documented, particularly in the organization theory literature. These are outward-facing boundary activities that help the team

deal with external players as sources of both critical resources and disruption. In contrast, boundary reinforcement represents a new and unstudied type of boundary work. This term encompasses the ways in which a team internally sets and reclaims its boundaries by increasing member awareness of boundaries and sharpening team identity. In other words, boundary reinforcement includes inward-facing boundary work that is focused on factors internal to the team. For example, the extent to which team members collectively create and maintain a clear identity and image can strengthen or weaken boundaries. In traditional hierarchical structures, each employee's tasks were prioritized with the assistance of an all-knowing supervisor or manager. This type of assistance, however, has become largely unavailable as functional chimneys have collapsed (Yan & Louis, 1999). In the newly reengineered boundary-less organizations, team members must juggle competing demands and face increasing pressure to draw a psychological boundary around them (Hirschhorn & Gilmore, 1992). Another challenge is to strengthen team boundaries to offset members' cross-functional differences and competing loyalties. In this way, the team can attract and hold members' attention on its central task. Dougherty (1992) noted that cross-functional teams consist of members with significantly different "thought worlds," each with its own fund of knowledge and system of meaning. Thus, team members are likely to gravitate naturally toward nonteam activities. Similarly, in her study of intraorganizational occupational boundaries, Bechky (2003) observed that different occupational groups continuously negotiate boundaries and frequently use organizational artifacts to reinforce their boundaries and maintain power and legitimacy. Thus, buffering the team from the demands of the environment may not be sufficient if there are no complementary activities that create a unique sense of belonging and identity for the team.

Conceptually, reinforcing team boundaries entails generating a sufficient and sustained centripetal or attracting force that stabilizes the perimeter of the team's space. The team must build a distinct identity and image to create a positive sense of belonging for members and demarcate its activity space from that of other organizational units. Increasing the salience of team boundaries reduces resource leakage, increases intrateam integration, and facilitates teamwork. To reinforce team boundaries, it helps to involve members in shaping the vision, increase cohesion by building team-specific symbols and language to keep the team's vision before members, honor homegrown rituals for making and celebrating progress, and adapt information and communication technologies to the team's situation (e.g., Beal, Cohen, Burke, & McLendon, 2003; Boland & Tenkasi, 1995; Sundstrom & Altman, 1989).

Boundary reinforcement is related to team performance in several ways. Salient boundaries enable the team to attract and preserve the energy and attachment of its members so that they can focus on carrying out the team's task. The enactment of the boundary may also benefit the team by increasing team identification and commitment. Finally, by keeping members' attention aligned with the team's mission and tasks, boundary reinforcement in knowledge teams may enhance individual and collective learning and creativity (Lovelace, Shapiro, & Weingart, 2001). On the basis of the above discussion, we propose the following hypothesis:

Hypothesis 3: Boundary reinforcement is positively associated with team performance and psychological safety.

Environment as a Moderator

A team's *environment* encompasses forces both within and beyond the larger organization that impinge on the team and its work. This fact accords with Ancona's (1990) argument for an external perspective on teams. *Task environment* refers to the set of entities that the team deals with directly in carrying out its mission, including suppliers and customers. Previous research suggests that team boundary work may be context dependent and task specific (e.g., Ancona & Caldwell, 1990). When forced to operate in different kinds of environments, teams may adopt different boundary management strategies or undertake different patterns of boundary actions. For example, a team in an uncertain environment may engage in boundary buffering and reinforcement more vigorously than it engages in spanning in order to attenuate environmental demands. Boundary researchers generally acknowledge a contingent quality to the relationship between boundary work and team performance (although this has yet to be empirically documented). For example, Choi (2002) argued for including task interdependence and environment as structural moderators. What is needed is a well-tested contingency view to guide teams in setting a strategy for boundary work. To that end, we propose and test two environmental moderators of the relationship between boundary work and performance. These moderators were chosen on the basis of their prevalence in past research, as well as their face validity or prevalence within the restructured organization.

Task uncertainty. According to the classic organizational literature, nonroutine tasks require complex problem solving, are highly uncertain, and cannot be accomplished using preset procedures (Galbraith, 1977; Van de Ven, Delbecq, & Koenig, 1976). By design, teams are most appropriate for accomplishing tasks that are complex, uncertain, and changing and have demanding information-processing requirements. We hypothesize that as task information-processing requirements become more onerous, the benefits associated with higher levels of coordination within and across the environment increase.

An increase in boundary spanning is to be expected in teams facing high levels of task uncertainty. By spanning, the team reaches out to map the environment more accurately, which, in turn, helps it better understand the new external demands and acquire needed information and resources. Boundary spanning in response to task uncertainty may take such forms as information gathering, network building, and general reconnaissance and mapping, as in the ambassadorial activities observed by Ancona and Caldwell (1992a).

In such conditions of high uncertainty, boundary buffering also has positive consequences, as it limits client design changes and time-consuming user inputs. This helps generate a clear and stable design, thereby producing a project that meets budget and schedule requirements. Similarly, boundary reinforcement actions can enhance the performance of uncertain tasks, because they allow team members to feel secure with their team and trust their teammates, thereby encouraging them to engage more freely in joint problem solving, knowledge sharing, and helping behaviors. Such processes have been shown to be important predictors of team effectiveness in accomplishing uncertain tasks (e.g., Edmondson, 1999;

Faraj & Sproull, 2000). On the basis of the above discussion, we propose the following hypothesis:

Hypothesis 4: Team boundary work in the form of (a) spanning, (b) buffering, and (c) reinforcement makes a greater contribution to team performance and psychological safety when task uncertainty is high compared with when it is low.

Resource scarcity. Resource scarcity is the extent to which available resources are not sufficient to support the sustained growth or survival of organizational entities (Aldrich, 1979; Pfeffer & Salancik, 1978). In contrast, slack, or abundance of resources, serves as a natural buffer during periods of crisis or environmental jolts. As organizations have striven in recent decades to be competitive, slack has been drastically cut (Kanter, 1989). Teams operating in such settings face the challenge of accomplishing the same amount of work or even more work with fewer resources, because cuts typically flow down and throughout the organization.

We expect that teams in a resource-scarce environment will be more actively engaged in boundary management than will teams operating in a resource-abundant environment (Choi, 2002). Acute need for additional resources can create a shared sense of urgency among team members and encourage them to look for both internal and external opportunities. Through spanning, teams seek to maintain current resource streams and secure new ones. Through buffering, teams block and control competing demands that threaten to produce bottlenecks in critical resources. Through boundary reinforcement, teams reemphasize or underscore in members' minds the primacy of the focal team and its task, leaving members free to apply their energy and commitment to the team's task and strengthening their resolve as they choose among competing calls on their attention. In contrast, when resources are relatively abundant, boundary work that is oriented toward seeking resources may be perceived as unnecessary, and effort spent on spanning boundaries may be seen as not adding much value. Thus, we suggest the following hypothesis:

Hypothesis 5: Team boundary work in the form of (a) spanning, (b) buffering, and (c) reinforcement makes a greater contribution to team performance and psychological safety when resource scarcity is high than when it is low.

Team Psychological Safety as a Mediator

Team psychological safety refers to the extent to which team members feel they can take interpersonal risks and speak their minds without being rejected or punished (Edmondson, 1999). *Psychological safety* describes a particular characteristic of a team at certain point in time and thus represents an emergent state of the team. As defined by Marks et al. (2001), "Emergent states describe cognitive, motivational, and affective states of teams," which "characterize properties of the team that are typically dynamic in nature and vary as a function of team context, inputs, processes, and outcomes" (p. 357). We follow these and other authors (e.g., S. G. Cohen & Bailey, 1997), who argued that emergent states are a function of team actions and processes and drive subsequent processes and outcomes. Furthermore, we posit that psychological safety mediates the relationship between team boundary activities and team performance.

We start from the assumption that a team's psychological safety is not a dispositional characteristic and does not emerge automatically. Rather, it is a psychological state that emerges as a result of team-building efforts, as well as other environmental factors. Hackman (2002) noted that psychological safety encourages members to jointly assess and collectively learn from task feedback. Edmondson (2002) found that organizational climates characterized by psychological safety encourage interpersonal risk taking and therefore lead to increased creativity and risk taking in teams. Similarly, Nembhard and Edmondson (2006) examined the effect of team structural factors on team psychological safety from the standpoint of power and social status differences. In a sample of teams in neonatal intensive care units, they found evidence that higher status members (based on a ranking of medical professions) are more likely to experience psychological safety and to subsequently engage in quality improvement efforts.

To date, researchers have tended to assume that the effect of psychological safety on team performance is indirect, expressed through innovation and learning impacts. However, the effect may be more direct for teams that must carry out complex knowledge tasks for which a climate of innovation is crucial. For example, psychological safety plays an important role in implementing improved practices in a medical setting (Tucker, Nembhard, & Edmondson, 2007). Similarly, according to an organizational analysis of 47 midsized German industrial firms, psychological safety is associated with innovativeness (Baer & Frese, 2003).

According to our model, team boundary activities help team members secure critical resources, manage information needs, and delineate and reinforce team boundaries so that team members are protected from environmental interference. These activities help them feel psychologically safe. More specifically, reinforced team boundaries help team members develop a shared sense of psychological safety (Edmondson, 1999). When team members routinely differentiate members from nonmembers, they develop a sense of belonging to the team, become tightly linked with each other socially, act sensitively in dealing with nonmembers, and likely experience a protected environment working within the team. Reinforced boundaries also help create a positive and mutually supportive climate, which enhances members' sense of satisfaction and willingness to stay on the team. Therefore, psychological safety can be seen as an intermediate outcome resulting from the team's effective management of its own boundaries.

Hypothesis 6: Team psychological safety mediates the relationship between boundary activities and team performance.

The hypotheses we have detailed concerning the relationships between boundary work and team performance, the moderating effects of the environment, and the mediating effect of psychological safety are depicted in the conceptual model in Figure 1. Next, we present the methods that we used in empirically testing the model.

Method

Sample

The teams that participated in this study were engaged in the development of application software for a variety of industries. All teams were employed by a leading information technology firm

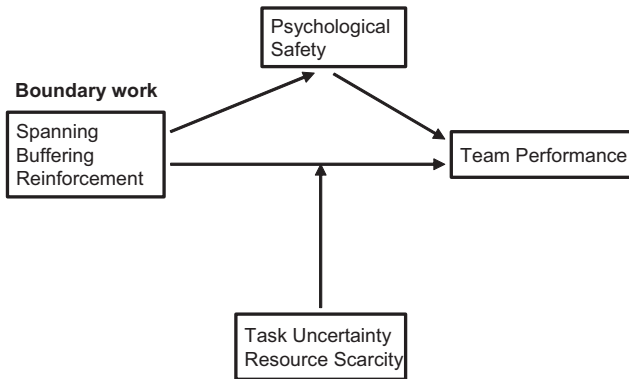


Figure 1. Model of boundary work in knowledge teams.

that specializes in developing business software for both external commercial customers and internal clients. With the assistance of a senior executive of the firm, we recruited teams using a set of criteria meant to mitigate confounding factors. Senior regional managers were contacted to nominate teams for the study. Participation was voluntary at both the team and the individual levels, and confidentiality was guaranteed. Respondents received pre-stamped envelopes for the return of completed surveys to a university address. To limit the impact of contextual and task factors, we selected teams (a) that reflected geographic diversity so that no single site was overrepresented; (b) that maximized the range of team history and reputation; (c) that were working on commercial application software, so as to maintain task comparability; (d) that had completed the requirements phase of the task life cycle, so as to ensure a well-defined task scope; (e) that were not working on projects that involved a significant number of external consultants; and (f) that had a size and duration of 4–15 members and 6–18 months, respectively, to minimize the impact of qualitatively different dynamics in large or long-lasting teams.

Two surveys were used for data collection. The main survey was mailed to each member of the 73 participating teams. A separate survey measuring different aspects of team performance was sent to two stakeholders for each team, both at the senior management level: one from the information systems (IS) group and one from the client side. These stakeholders were identified by the team leaders, and none of the stakeholders were direct members of the team to which they reported.

Responses were received from 64 out of the 73 teams whose participation was solicited for the study (a team-level response rate of 88%), representing 12 key company sites across the United States, including both regional centers and independent sites.¹ A total of 290 individual team members participated in the study. The average team size was 10, and, on average, 59% of team members responded. The intrateam response rate ranged from 25% to 100%. Teams with fewer than 2 respondents were dropped from the analysis. Our sample included 4 teams with only 2 respondents, 12 teams with 3 respondents, 18 teams with 4 respondents, and 30 teams with 5 or more respondents. In addition, performance ratings of these teams were collected from 120 stakeholder respondents. Statistical tests indicated that there were no significant differences in key variables among sites.

On average, respondents were 39 years of age and had 12 years of experience in the software field, of which they had spent 11

years employed by the company. With regard to education, 86% of the respondents had completed college and 23% had a master's or higher degree. The organizational roles performed by our respondents included team leaders (20%), programmers (29%), system analysts (18%), specialists or consultants (16%), and other professionals (13%). Thirty-six percent of the respondents were women.

Measures

Measures of the variables were pretested with 40 software development professionals participating in a masters of science program in management information systems. These graduate students possessed multiple years of experience in software development and worked in team environments comparable to those of the target population. The pretest group commented on the clarity, format, and content of the survey items. We conducted follow-up interviews with some pretest participants and revised and refined the survey items on the basis of their insights. All items were assessed on a 5-point Likert-type scale.

Because our model focuses on the team level, we took a number of steps before aggregating individual responses to the team level. First, to ensure that the level of measurement matches the level of theory, all survey items were worded to reference the team environment and activities. Second, to justify aggregation, we computed an interrater agreement statistic using the r_{wg} procedure (James, Demaree, & Wolf, 1984) to assess the convergence of responses among team members. The median r_{wg} values ranged from .85 to .94, above the generally accepted level of .7, indicating strong agreement (Bliese, 2000; Castro, 2002). Third, we report intraclass correlations (ICCs) to test whether the measures are reliable enough to be aggregated to the team level (Bliese, 2000; Chen, Mathieu, & Bliese, 2004). The ICC₁ calculates the percentage of variance in a variable that is attributable to team membership, and the ICC₂ indicates whether teams can be reliably differentiated on the basis of average member ratings. Fourth, we performed a one-way analysis of variance on each variable to assess whether interteam variance was larger than intrateam variance. All of our variables were significant at the $p < .05$ level (except for buffering that was borderline, $p = .07$). Finally, to ensure internal consistency of measurement, we calculated the Cronbach's alphas for all the variables at the individual ($\alpha_{\text{individual}}$ range = .73 to .85) and team levels ($\alpha_{\text{aggregate}}$ range = .79 to .91). On the basis of these results, individual responses were aggregated and data were analyzed at the team level.

Team performance. In this study, team performance measures the extent to which the team achieves its project goals. We used stakeholder performance ratings for this study because more objective measures of performance, such as lines of code per person month, were unavailable and are, in any case, recognized as problematic and subject to manipulation (Henderson & Lee, 1992). Previous scholars have suggested that compared with traditional production teams, the performance of knowledge workers, such as software development teams, is often multidimensional and thus hard to assess appropriately (Ancona & Caldwell, 1992a; Faraj &

¹ Four teams did not complete the study because they changed managers during data collection. Five other teams had their projects put on hold or cancelled.

Sproull, 2000). For example, the members of the organization of which the team is a part may adopt different performance criteria than do the external stakeholders. Therefore, for this study, we collected performance ratings from both the software development organization and the client organization to generate a multisource assessment of performance and avoid institutional bias. We gathered ratings from senior IS managers in the firm who were familiar with the project teams' work but were not involved on a day-to-day basis. From the client side, the project-sponsoring manager provided his or her assessment. We adapted questions used and validated in a similar context by Henderson and Lee (1992) and Guinan, Coopridge, and Faraj (1998). The four questions focused on how well the team performed compared with other software teams with which the stakeholders were familiar, with regard to the following areas: adherence to schedules, adherence to budgets, ability to meet project goals, and the extent to which users are satisfied with the system. We performed a principal-components analysis (all five items loaded onto a single dimension) to confirm that stakeholders perceived performance holistically. A correlation analysis ($r = .29$, $p < .03$) supported our view that both sets of stakeholders had parallel assessments of the team. As a result, we aggregated the four items to develop a measure of team performance: $\alpha_{\text{individual level}} = .85$; $\alpha_{\text{aggregate level}} = .84$; median $r_{\text{wg}} = .86$; $\text{ICC}_1 = .1$; $\text{ICC}_2 = .34$.

Boundary work. Measurement of boundary spanning was based on four survey questions that assessed the extent to which the team encourages its members to solicit critical information and resources, influence important external agents or actors, and build relational networks: $\alpha_{\text{individual level}} = .73$; $\alpha_{\text{aggregate level}} = .81$; median $r_{\text{wg}} = .87$; $\text{ICC}_1 = .17$; $\text{ICC}_2 = .50$. Boundary buffering was measured by four items that assess the extent to which the team and its leadership deflect or absorb external interference and competing demands or requests and how well they prevent team members from overloading: $\alpha_{\text{individual level}} = .79$; $\alpha_{\text{aggregate level}} = .82$; median $r_{\text{wg}} = .85$; $\text{ICC}_1 = .06$; $\text{ICC}_2 = .24$. Finally, boundary reinforcement was measured by another four survey items, which assessed the team's effort to create a sense of boundary awareness among members and build a clear and distinctive team identity: $\alpha_{\text{individual level}} = .83$; $\alpha_{\text{aggregate level}} = .84$; median $r_{\text{wg}} = .92$; $\text{ICC}_1 = .17$; $\text{ICC}_2 = .49$. The boundary work variables were analyzed using a principal-components factor analysis with an Oblimin rotation. The three factors that emerged replicated the planned scales and accounted for 63.5% of overall data variance. All items loaded higher than .61 on their core dimension, and all cross-loadings were below .32. Table 1 shows the full set of items underlying each of the three boundary work variables.

Task uncertainty. Measurement of task uncertainty depends on how effectively the work of transforming inputs into outputs can be reduced to well-defined operational steps, procedures, and routines. The instrument developed by Whitney, Daft, and Cooper (1983) has been used in a large number of studies, and its reliability and validity have been confirmed. We used the four-item version validated by Nidumolu (1995) in a similar software development setting: $\alpha_{\text{individual level}} = .79$; $\alpha_{\text{aggregate level}} = .86$; median $r_{\text{wg}} = .88$; $\text{ICC}_1 = .2$; $\text{ICC}_2 = .54$.

Resource scarcity. Dess and Beard's (1984) operationalization of resource abundance, which is rooted and tested in the organizational context, served as a basis for our measurement of resource scarcity (coded in the opposite direction) at the team

level. The three reverse-coded items for this variable measured the extent to which the team (a) finds it critical to preserve and stretch its available resources to accomplish its task, (b) has to carry out its tasks under serious resource constraints, and (c) experiences an ongoing need for additional resources to get its job done: $\alpha_{\text{individual level}} = .72$; $\alpha_{\text{aggregate level}} = .79$; median $r_{\text{wg}} = .85$; $\text{ICC}_1 = .26$; $\text{ICC}_2 = .62$. In a principal-components analysis, the two expected factors emerged, which confirmed that task uncertainty and resource scarcity are conceptually distinct.

Psychological safety. Data for measuring team psychological safety were collected from team members using a six-item instrument developed and tested by Edmondson (1999). Sample items include "Mistakes are held against you as a member of this team" (reverse coded) and "It is OK for team members to talk about competing demands and how they handle them": $\alpha_{\text{individual level}} = .85$; $\alpha_{\text{aggregate level}} = .91$; median $r_{\text{wg}} = .88$; $\text{ICC}_1 = .19$; $\text{ICC}_2 = .52$.

Results

Table 2 presents the means, standard deviations, Cronbach's α , and zero-order correlations for the variables in the study. Before testing the hypotheses, we developed statistical power calculations to ascertain whether our team-level analysis faces sample-size limitations. Using J. Cohen's (1988) benchmarks (i.e., the level of power of .8 and an R^2 level of .26), which are appropriate for organizational and social psychological studies and in line with similar team-level studies (Ancona & Caldwell, 1992a; Nidumolu, 1995; Smith et al., 1994), we were required to have a sample of 61 teams. Because our actual sample contained 64 teams, the sample-size requirement was adequately met.

Hypothesis Testing

Because our theoretical propositions required the testing of both direct links of boundary work variables as well as moderating and mediating effects on team performance, hierarchical regression analyses were conducted. Variables were mean centered to reduce the potential problem of multicollinearity (J. Cohen, Cohen, West, & Aiken, 2003). The results of our analyses are reported in Table 3. To confirm that multicollinearity is not present, we examined the variance inflation factor (VIF) and the condition index (CI) of each independent variable. All VIF values were less than 1.9, whereas the highest CI was 2.6. Both are well below the warning levels (i.e., a VIF value of 10 and a CI value of 30) that indicate multicollinearity problems (J. Cohen, Cohen, West, & Aiken, 2003, p. 424).

Psychological safety. Model 1 examines the direct relationship between each boundary work variable (buffering, spanning, and reinforcement) and psychological safety. The model is significant overall, $R^2 = 34.8\%$, $p < .001$, and thus supports our general proposition that boundary work is positively linked to team psychological safety. Both boundary buffering ($p < .01$) and boundary reinforcement ($p < .05$) exhibited significant unique effects related to psychological safety, thus supporting Hypotheses 2 and 3, respectively. This shows that the chief purpose of buffering boundaries is noise reduction, whereas the chief purpose of reinforcing boundaries is internal order maintenance. As extraneous competing demands are blocked out and internal awareness of boundaries is enhanced, the team's internal coherence,

Table 1
Oblimin Factor Loadings for Boundary Work (Individual Level of Analysis)

Item	Factor 1	Factor 2	Factor 3
Boundary reinforcement			
To what extent has this team tried to create a clear sense of its identity and purpose?	.88		
To what extent does this team's image of itself and its task distinguish it from other teams in the division?	.82		
To what extent is this team's image clear to important outsiders with whom team members interact?	.79		
To what extent do members share a common understanding of the team's image or identity?	.68		
Boundary buffering			
To what extent are outside pressures deflected or absorbed so that the team can work free of interference?		.90	
To what extent are outsiders prevented from "overloading" the team with either too much information or too many requests?		.89	
To what extent does the team or team leader help team members manage demands placed on them by other organizational units?		.67	
To what extent do you feel that team members work in a well-buffered or protected environment?		.61	
Boundary spanning			
To what extent does the team encourage its members to solicit information and resources from elsewhere in and/or beyond the division?			.83
To what extent does the team encourage its members to try to influence important actors elsewhere in and/or beyond the division on behalf of the team and its work?			.72
To what extent does the team value team members for making use of their relationships with others on behalf of the team?		.32	.66
To what extent does the work of the team depend upon information and resources actively solicited by team members, that is, information and resources beyond what comes through official channels?			.64
Eigenvalues	4.47	1.90	1.25

Note. Loadings smaller than .3 are not shown.

attractiveness, and member satisfaction are strengthened. For boundary spanning, the relationship is positively signed but statistically insignificant. Task uncertainty and resource scarcity were added to Model 2 to assess their direct effect. The results show significant explanatory impact ($\Delta R^2 = .12$, F for $\Delta R^2 = 6.28$, $p < .01$). In Model 3, all six interaction terms were entered to test for the hypothesized moderating effects. This model provides significant explanatory power, compared with Model 2 ($\Delta R^2 = .1$, F for $\Delta R^2 = 2$, $p < .05$). Specifically, the proposed moderating effect of task uncertainty on the relationship between buffering and psychological safety and the moderating effect of resource scarcity on the relationship between boundary reinforcement and psychological safety are both significant, thus rendering support for Hypotheses 4b and 5c, respectively.

Team performance. The same procedure of hierarchical regression was repeated for team performance, the other dependent variable. Boundary reinforcement ($p < .05$) was found to be positively linked to team performance, suggesting a direct relationship between the two variables. This adds further support for Hypothesis 3. In testing moderating effects, significant results were generated for four out of the six interaction terms, suggesting that task uncertainty and resource scarcity each consistently moderate the relationship between boundary spanning and team performance and between boundary reinforcement and team performance. These results render strong support for our Hypotheses 4a, 4c, 5a, and 5c.

Following J. Cohen et al. (2003), the above reported interaction effects are graphically plotted in a series of figures. Figure 2A

Table 2
Means, Standard Deviations, Reliabilities, and Correlations Among Variables

Variable	<i>M</i>	<i>SD</i>	α (individual level)	α (aggregate level)	R_{wg} (median)	Correlation coefficient						
						1	2	3	4	5	6	7
1. Boundary spanning	3.17	0.49	.73	.81	.79	—						
2. Boundary buffering	3.22	0.44	.79	.82	.79	.28	—					
3. Boundary reinforcement	3.53	0.47	.83	.84	.88	.63**	.34**	—				
4. Task uncertainty	2.76	0.54	.79	.86	.81	-.26*	-.42*	-.37**	—			
5. Resource scarcity	3.10	0.56	.72	.79	.83	.32**	-.02	.18	.01	—		
6. Psychological Safety	3.98	0.48	.85	.91	.83	.38**	.5**	.45**	-.33**	-.25*	—	
7. Team performance	4.03	0.55	.85	.84	.86	.05	.24*	.23	-.33**	-.27*	.23	—

Note. All variables were assessed on a 5-point scale. Item anchors for team performance were 1 = *well below average* and 5 = *well above average*. All other item anchors were 1 = *to a very small extent* and 5 = *to a very great extent*. R_{wg} for team performance measures the extent of agreement among stakeholders.

* $p < .05$. ** $p < .01$.

Table 3
Regression Results for Psychological Safety and Team Performance

Variable	Psychological safety			Team performance			Mediation
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
Main effects							
Boundary spanning	.11	.25*	.23*	-.19	-.09	-.09	-.07
Boundary buffering	.39**	.33*	.35**	.19	.09	.19	.21
Boundary reinforcement	.25*	.23*	.35**	.29*	.22	.26	.28
Task uncertainty		-.04	.02		-.23*	-.20	-.20
Resource scarcity		-.36**	-.37**		-.27*	-.29**	-.32*
Task Uncertainty $\times$ Spanning			.06			-.46	-.45
Task Uncertainty $\times$ Buffering			.20*			.02	.03
Task Uncertainty $\times$ Reinforcement			-.10			.50*	.49
Resource Scarcity $\times$ Spanning			-.07			-.66**	-.67**
Resource Scarcity $\times$ Buffering			.01			-.08	-.08
Resource Scarcity $\times$ Reinforcement			.29*			.64**	.66**
Psychological safety							-.07
Model statistics:							
N	64	64	64	64	64	64	64
R ²	34.8%	46.4%	56.4%	10.3%	21.4%	36.6%	36.8%
Adjusted R ²	31.5%	41.7%	47.3%	5.8%	14.7%	23.2%	21.9%
Model F	10.7**	10**	6.1**	2.3*	3.2**	2.7**	2.47*
Change in R ²		.12	.10		.11	.15	.02
F for change in R ²		6.3**	2*		4.1*	2.1*	.16

* $p < .05$. ** $p < .01$ (one-tailed).

indicates that the relationship between buffering and psychological safety is positive only under conditions of high task uncertainty, suggesting that when team tasks are ill defined and uncertain, efforts to reduce environmental noise and external intervention enhance team members' sense of psychological safety. Figure 2B points out that the relationship between boundary reinforcement and psychological safety is significant only under conditions of high resource scarcity. This suggests that when a team must operate in a resource-lean environment, reinforcing its boundaries helps it build psychologically safer turf for its members. In Figure 2C, the relationship between boundary reinforcement and team performance is positive under conditions of high task uncertainty and negative under conditions of low task uncertainty. This suggests that task uncertainty plays a dual role in moderating the relationship between boundary reinforcement and team performance. When the team's task is uncertain, reinforcing boundaries will help it achieve its project goals. However, when the team performs a task that is well defined and without uncertainty, boundary reinforcement may be unnecessary, adding no value and wasting team resources, which hinders performance. Figure 2D also shows that boundary spanning is positively related to team performance under conditions of low resource scarcity but negatively related to performance under conditions of high resource scarcity, suggesting that in a resource-abundant environment, performance is enhanced through spanning. However, when few resources are available, spanning may consume team effort to the detriment of the team's task. This finding is somewhat unexpected and is further elaborated in the next section.

Results for mediation effect of psychological safety. The output from Model 7 in Table 3 indicates whether psychological safety mediates between boundary work and team performance. The hypothesis was tested to confirm the previously established link between psychological safety and learning activities in teams

(Edmondson, 1999), as well as the links between psychological safety and innovation performance (Baer & Frese, 2003) and between psychological safety and medical team performance (Tucker et al., 2007). We evaluated the model by entering psychological safety as a last step in the hierarchical regression framework used to test the previous models. We used the logic of mediation analysis (Baron & Kenny, 1986). All three boundary work variables were positively correlated with psychological safety ($r_{\text{spanning}} = .38, p < .01$; $r_{\text{buffering}} = .5, p < .01$; $r_{\text{reinforcement}} = .45, p < .01$). Psychological safety was weakly correlated with team performance ($r = .23, p < .1$). Model 7 (see Table 3) provides the last step to establish mediation. The results indicate that psychological safety was not a significant mediator between boundary activities and team performance (Model $F = 2.47, p < .05$; $\Delta R^2 = .02, ns$). Thus, Hypothesis 6 was not supported.

Additional analysis. We also tested a three-way interaction model to establish whether the moderating effects of task uncertainty and resource scarcity operated jointly. Here too we used our hierarchical analysis framework to construct a model in which the three-way terms (e.g., Task Uncertainty $\times$ Resource Scarcity $\times$ Spanning) were introduced as a last step. The results were not significant for either psychological safety ($\Delta R^2 = .01, ns$) or team performance ($\Delta R^2 = .005, ns$). One possible explanation for the lack of three-way interaction results could be the lack of statistical power due to limited sample size. Together, these additional analyses provide further confidence in the relevance of our theoretical model.

Discussion and Conclusions

Our primary goal in this study is to develop and test a model of the consequences of boundary work in knowledge teams. Our empirical results are generally supportive of the theoretical model we propose regarding the direct relationship between boundary

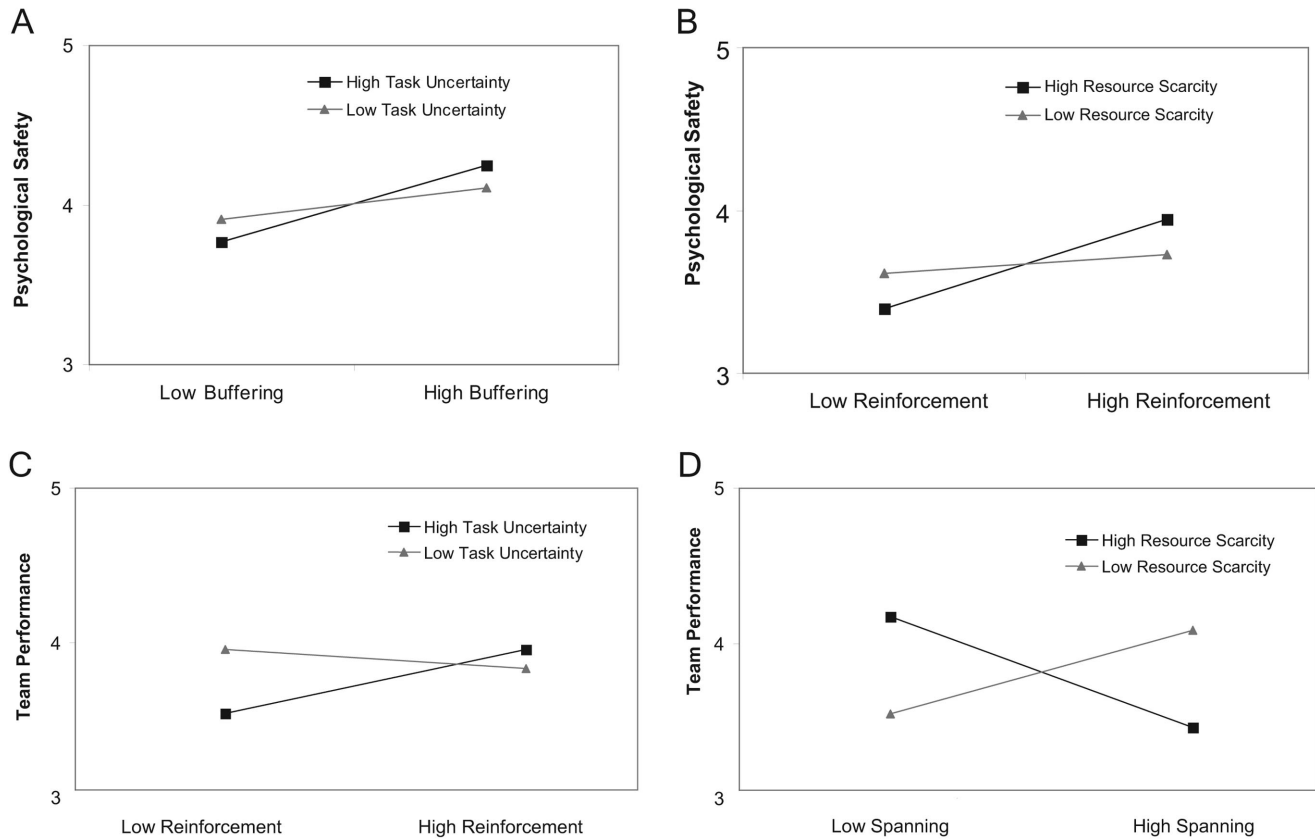


Figure 2. Moderating effects. A: Effect on psychological safety of the interaction between task uncertainty and boundary buffering. B: Effect on psychological safety of the interaction between resource scarcity and boundary reinforcement. C: Effect on team performance of the interaction between task uncertainty and boundary reinforcement. D: Effect on team performance of the interaction between resource scarcity and boundary spanning.

work on the one hand and team performance and psychological safety on the other. The results also support the hypothesis that task uncertainty and resource scarcity have moderating effects on the relationship between boundary work and team performance, as well as on the relationship between boundary work and psychological safety. The two full models explain a large percentage of the variance (36.6% and 56.4%, respectively) in team performance and psychological safety. All five hypotheses receive partial or full support.

These findings are important for three reasons. First, we provide empirical validation for a boundary activities framework that differentiates between outward-facing boundary activities, such as buffering and spanning, and inwardly focused activities, such as boundary reinforcement. Second, we demonstrate the importance of boundary activities in knowledge teams engaged in software development, a task qualitatively different from those performed by R&D and new product development teams, which were the focus of previous research (e.g., Ancona & Caldwell, 1992a). Third, we show that the link between boundary work and team performance is moderated by environmental factors, supporting Choi's (2002) call for a new theory to explain the impact of moderating factors on team boundary activities.

Consider the direct relationship between boundary work and team performance. Collectively, our model results support the

contention that the three dimensions of boundary work contribute to team performance. Individually, such a direct effect was found for each of the three types of boundary work in different regression models. More specifically, our findings offer clear empirical support for the necessity of boundary reinforcement, that is, inwardly focused boundary work. Boundary reinforcement contributes to both an emergent social state (psychological safety) and the team's economic outcomes (team performance), underscoring the importance of inwardly focused boundary work in the team setting. This result enriches the literature on organizational teams, which has, until now, characterized boundary work as activities directed toward agents and players external to the focal team while deeming actions undertaken toward internal players insignificant or even irrelevant to team boundaries (e.g., Ancona, 1990; Tushman, 1977).

Our findings also highlight the importance of a team's task and organizational context. Both task uncertainty and resource scarcity played significant roles in moderating the relationship between boundary work on the one hand and team performance and psychological safety on the other. In the full regression model of team performance, four out of the six interaction terms were significant, indicating the strength and scale of the moderating effect. This evidence further strengthens the open systems approach to organizational teams and suggests that the team's organizational con-

text can dictate the choice of boundary work strategies. Specifically, we found that boundary buffering contributes to team psychological safety when a team's tasks are uncertain, and boundary spanning helps the team achieve its goals when resources are abundant but hinders goal achievement when resources are scarce. These results corroborate Ancona and Caldwell's (1988) findings regarding the contradictory impact of scouting and external ambassadorial activities. Boundary reinforcement is positively linked to team performance under conditions of high task uncertainty and high resource scarcity, which confirms other findings as to the importance of resources for successful completion of complex technical projects (Baba, 1995).

Another important finding was that team boundary reinforcement appears to be necessary for the creation of an interpersonal safe haven within the team (Edmondson, 1999). Thus, our findings corroborate the dominant perspective on psychological safety—that it is an emergent state that plays an important role in team performance, although its impact may be manifested primarily through more specific processes (Ilgen et al., 2005; Marks et al., 2001). Although recent research has found links between psychological safety and performance in medical care units (Tucker et al., 2007) and small industrial firms (Baer & Frese, 2003), it is possible that there are differences due to different levels of analysis. Our results are more in line with findings at the team level, where the impact of psychological safety is deemed important but is believed to occur through such processes as learning (e.g., Edmondson, 1999) and goal-directed task work (Marks et al., 2001).

In conclusion, consistent with earlier work on R&D teams (e.g., Allen, 1984; Tushman, 1977) and product development teams (e.g., Ancona & Caldwell, 1992a; Brown & Eisenhardt, 1997), this study confirms the importance of team-level boundary work for knowledge task teams. It also provides empirical support for boundary reinforcement, a new and distinct form of boundary work that complements the outwardly focused boundary work of buffering and spanning.

Hypotheses Receiving No or Partial Support

Our hypotheses regarding boundary spanning received only partial support, a finding that deserves elaboration. The effect of spanning on psychological safety was not significant in Model 1, the simplest regression model. It becomes significant in Model 2 when the environmental moderators are included. Finally, in Model 3, spanning did not interact with those same moderators. This suggests that the effect of boundary spanning on psychological safety is far from direct or simple but occurs only in certain organizational contexts or in the presence of certain environmental factors. This result points to differences between the closed system and the open systems mindsets. In a closed system environment, boundary spanning activities may not be perceived as anything important or even relevant to team members. However, when certain environmental contexts, such as task complexity and resource scarcity, are taken into account, boundary spanning increases in importance to the team and positively affects the emerging psychological states of the team members.

Spanning has no direct effect on team performance but is negatively linked to team performance under conditions of resource scarcity. This suggests that in an environment where re-

sources are relatively abundant, performance is enhanced through spanning. However, in resource-poor environments, additional boundary spanning may actually hinder performance. These results are at odds with Ancona and Caldwell's (1988) interview-based findings regarding the efficacy of spanning-type efforts by ambassadors and scouts among new product development teams. The apparent contradiction is most likely due to the differences in the types of tasks the teams perform, as spanning may be highly sensitive to the organizational context in which the team is found. In new product development, boundary spanning is likely to be crucial throughout the task life cycle, whereas software development benefits more from spanning at earlier stages of the task, such as when user requirements are elicited and the design is reviewed. During the later, main phase of software development, the team's efforts are internally focused on coding (writing computer instructions) and software testing. Not surprisingly, spending time on boundary spanning when the task at hand is writing and testing code may be detrimental.

In the current literature, the notion of spanning serves multiple purposes (e.g., Ancona & Caldwell, 1992b). It involves the acquisition of outside resources and outreach to achieve team self-promotion and network building. On the one hand, spanning activities help generate information and resources, which leads to improved team task performance. On the other hand, these activities expend effort, energy, and time; thus, team resources are dispersed and attention diluted, with negative consequences for performance. These contradictory outcomes are in line with the findings of Marrone et al. (2007) that the benefits of increased spanning come at the cost of member overload. Our findings seem to indicate that in an environment where resources are relatively abundant, performance is enhanced by networking with and thus influencing external stakeholders. However, in teams facing resource scarcity, spanning activities for the purposes of self-promotion and network building may be perceived as competing for the same pool of resources needed to carry out the task at hand. Therefore, this unnecessary spanning activity is counterproductive and hinders functions aimed directly at achieving the team's goals.

Our Hypothesis 6 regarding the mediating effect of psychological safety did not receive support (see Model 7 in Table 3). There are several plausible explanations for the lack of support for this hypothesis. First, although psychological safety as an emergent state may play an important role in team performance, its impact may be indirect and manifest itself through other emergent states or team processes (Ilgen et al., 2005; Marks et al., 2001). For example, psychological safety created through team boundary work increases learning (Edmondson, 1999) and may positively impact cohesion and coordination (Marks et al., 2001). Although these subsequent emerging states and processes can lead to performance, the impact of psychological safety on performance may remain indirect. Second, although recent research has found links between psychological safety and performance in medical care units (Tucker et al., 2007) and small industrial firms (Baer & Frese, 2003), it is possible that the differences are due to the different level of analysis (larger and more complex work units) in those studies. Overall, our findings in the software development context do seem to support the notion that psychological safety is an emergent state whose impact on performance needs to occur via specific team processes.

Theoretical and Practical Implications

This study contributes to the literature in several ways. First, in our conceptual model, we adopt a holistic approach to encompass both the outwardly and the inwardly oriented dimensions of boundary work. In recent studies that adopted an open systems perspective, emphasis has been placed on how teams reach out to the environment. Meanwhile, the process by which teams work internally to tend their boundaries has received little attention. Therefore, our findings regarding boundary reinforcement help fill the gap between team process research (e.g., Marks et al., 2001) and the open systems perspective on teams (e.g., Ancona & Caldwell, 1992a), thus integrating these two streams of research.

Second, in response to previous scholars' calls for studies of moderating factors in team boundary research (Choi, 2002), we have theorized and empirically validated a model of how the relationship between boundary work and team effectiveness is moderated by team task uncertainty and resource scarcity. Empirical evidence confirms this contingency approach, thereby helping enrich previous studies by adding explanatory power to the complex functions of team boundary work.

Third, our findings appear to indicate that previous research on team boundary management may be context dependent and task specific. We conducted this empirical study in a parsimonious, single-organization setting. We demonstrated the importance of boundary work for knowledge teams engaged in software development, a knowledge task qualitatively different from those carried out by R&D and new product development teams, which have been the primary focus of previous research.

Fourth, we measured team performance from the perspectives of different stakeholders. This approach adds robustness to performance measurement and contributes to a multidimensional, multistakeholder approach to team effectiveness.

The results of this study can help practitioners identify conditions under which boundary spanning, buffering, and reinforcement make sense. We offer empirical support for Choi's (2002) theorizing regarding the trade-offs in time and effort required to manage the competing demands of a team's internal and external processes. Team leaders need to monitor critical features of the environment in which they work, and they must use the information they gather to determine which boundary management strategy they should adopt. Management practitioners can enhance team effectiveness by going beyond internal team design factors and recognizing the importance of helping a team manage its boundaries. Specifically, team members need to be sensitive to the costs and benefits associated with boundary work. Clearly, if a team devotes more attention to external activities, it has less time to devote to internal activities. Thus, boundary work should be carefully designed so that both the costs incurred and the benefits derived from it are taken into account.

Finally, it may be important for managers to recognize the need to support team efforts to reinforce their boundaries when the organizational environment is highly fluid and members face competing demands. Reinforcing boundaries may be beneficial when teams must build or refine their own identities and subcultures. However, there are drawbacks; for example, competition between teams may give rise to cultural clashes and turf-defending wars, as Bechky (2003) observed in her study of occupational teams. This poses a challenge for the corporate-level management from a

structure and incentives standpoint. Empowering these diversified teams and achieving alignment between them is a challenging task.

Limitations and Future Research

There are several limitations to this study, which should be noted to help future researchers. First, the present study was conducted in a relatively homogeneous environment. All teams were drawn from the same organization and were engaged in comparable tasks. As a result, the range of responses regarding measures of environmental moderators was possibly too narrow to represent a sufficient test of the model. Future research should encompass more heterogeneous samples of teams and team environments and should examine the efficacy of team boundary work in other knowledge domains. A second limitation is potential bias in measuring team performance. As noted by Sundstrom, DeMeuse, and Futrell (1990) and others, there is no optimal way to measure team performance. However, collecting data from multiple team members and stakeholders and using multi-item measures should attenuate these concerns. Future research could benefit from the collection of both quantitative and qualitative measures of performance. Finally, the design of this study does not allow for causal interpretation; instead, it provides a test of the association or relationship between the variables. Both longitudinal designs and in-depth qualitative inquiries are needed to examine the causal relationships implied by our model.

Regarding potential directions for future research, our findings on boundary reinforcement suggest that greater clarity is needed with respect to what reinforcement entails and how it differs from traditional team development activities. Boundary work has been shown to be critical for teams engaged in R&D and new product development, and our findings suggest that these activities may be necessary for other types of teams as well. Future researchers could also investigate the relevance of boundary work outside the domain of software development.

Conclusion

Researchers and practitioners alike have grappled with the implications of the emerging postbureaucratic organizations, in which teams are a primary organizing structure. As teams become increasingly central work structures, it is becoming necessary to devise ways to ensure that pivotal work functions are accomplished and team members are protected from external demands. As teams face increasingly demanding institutional environments, they must respond to institutional forces to gain legitimacy, secure sponsorship or support, and otherwise provide for team survival (DiMaggio & Powell, 1983). This study shows the importance of team boundary work, links specific boundary work activities to team performance and psychological safety, and establishes important task and environmental contingencies. What are the consequences of neglecting such boundary functions? How are they attended to, by whom, and with what means? Answers to these questions will have important theoretical and practical implications.

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